Matheus Fonseca

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Experience

CodePath Remote
Tech Fellow | AI110 (Applied AI Engineering) and TIP102 (Intermediate Technical Interview Prep) Feb 2026 – Present

- Supported instruction for nearly 1,000 students across two courses, driving engagement and comprehension through interactive facilitation.
- Assisted with weekly sessions covering data structures, algorithms, and AI literacy.
- Mentored students on technical interview preparation, focusing on coding strategies and writing clean, efficient code.
- Broke down complex DSA and AI concepts into clear, accessible explanations for students with varying technical backgrounds.

AlgoVerse Remote
AI Researcher Aug 2025 – Present

- Researching model merging techniques to combine complementary LLMs for improved adversarial robustness.
- Built pipeline to assess pairwise disjointness across transformer architectures and identify complementary model pairs.
- Exploring merging methods for models with similar parameter counts and layer structures.
- Working toward peer-reviewed publication in machine learning.

Cava Group, Inc. New York, NY
Team Member Apr 2025 – Sep 2025

- Demonstrated adaptability across multiple fast-paced roles including line service, grill, prep, and dishwashing.
- Consistently recognized by management for exceptional work ethic, curiosity, and high-quality customer service.
- Recognized for promotion to managerial position but chose to leave to prioritize studies.

Honors / Awards

Morgan Stanley CodeToGive Hackathon | 1st Place, Track A Mar 2026

- Selected as one of 100 from 2,200+ applicants
- Built a self-serve volunteer flyering platform for Lemontree, a non-profit providing free food to those in need, enabling volunteers to run campaigns without staff involvement.
- Drove core ideation and developed Citrus, an AI agent that guides volunteers through the flyering process in real time.
- Built leaderboard, get-started page, and onboarding guide; implemented nearby printer functionality and mobile UI optimizations and animations.

Projects

SendToMac | python, supabase, rumps, swift

- Developed a **cross-platform application** enabling asynchronous transfer of text and URLs from iOS to macOS, even while the Mac is offline.
- Implemented **background macOS client** using **Python**, **AppKit**, and **rumps** to display real-time alerts with user actions (Open, Snooze, Ignore).
- Prototyped iOS interface using **SwiftUI** with AI assistance to accelerate design and implementation.
- Enabled content transfer via **REST API** with responsive success/failure alerts.
- Integrated scheduling and expiration logic using timestamps to manage message flow and cleanup.

Balatest (backend) | python, flask, pypdf2, gemini, prompt engineering

- Led a 4-person team from ideation to completion at **CCNY's Byte Hacks**, delivering a playable educational card game prototype in 24h earning positive feedback from judges/participants.
- Engineered **backend service** in **Flask** to parse PDF content with PyPDF2 and dynamically generate JSON-based playing cards across multiple difficulty levels.
- Integrated Gemini API with prompt engineering techniques (Chain-of-Thought–inspired reasoning, structured output constraints) for semantic processing and adaptive gameplay content generation.
- Collaborated on game logic and design to align backend functionality with front-end mechanics and user experience.

Chess Web-App | html, css, javascript

- Developed an interactive **chess web application** with full rules enforcement, including check, checkmate, stalemate, and legal move validation.
- Implemented real-time piece selection with visual highlights and move restrictions based on king safety and game rules.
- Built modular move logic and game state tracking to ensure accurate gameplay flow and turn handling.
- Designed a responsive, user-friendly interface using **HTML**, **CSS**, and **JavaScript**.

Education

CUNY Brooklyn College Expected 2029
Bachelor of Science, Major in ***Computer Science*** GPA: 3.95 / 4.0

- **Relevant Coursework:** Intro to Programming (**Java**), **Calculus I** *(Spring 2026)*, **Discrete Structures** *(Spring 2026)*, **Modern Programming Techniques** *(Spring 2026)*

CS50x: Introduction to Computer Science – Harvard/edX Dec 2024

- DSA and memory management in **C**; web development in **Python**, **JavaScript**, **SQLite**, and **HTML/CSS**.

CodePath – TIP102 (Advanced Track) Nov 2025

- Selective program covering **data structures**, **algorithms**, and problem-solving patterns through 50+ coding problems.

Skills

Technical: Python, Flask, Java, C, JavaScript, CSS/HTML, SQLite, Supabase, Git, REST APIs
Spoken Languages: English (Native), Portuguese (Native), Spanish (Basic)
Interests: Chess, Weightlifting, Swimming, Travel, AI, Movies